Evaluation of an Immersive Audio Augmented Reality Device for the Exploration of Ancient Ruins

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Abstract

The use of audio augmented reality (AAR) devices which improves the auditory immersion of visitors can be relevant in museum settings. This study evaluates a soundwalk apparatus which combines geolocation, head tracking, and virtual acoustic simulations developed for the exploration of Gallo-Roman theater ruins. The soundwalk consisted in a 25-minute sound production which was experienced by 343 participants on site. Their responses to a questionnaire helped to prioritize future developments for AAR experiences. Results showed that the narrative engagement is the key element, followed by the ability to forget the technical setup and the capacity to externalize virtual sounds.

Keywords: Geolocated Soundwalks, Immersion, Acoustic Simulation, Spatial Audio, Archaeoacoustics, Audio Augmented Reality

1 Introduction

AAR is defined as "the enhancement of reality through virtual sounds" [4] and aims to overlay a plausible virtual audio environment onto reality. The use of geolocated devices transforms the visitor experience into a more sensory and personalized one [2]. Inclusion of head tracking and virtual acoustic simulation is expected to further promote the visitor immersion, by anchoring virtual sound sources around the listener in a real environment [10]. In order to measure the sensation of immersion, multiple criteria defined by Lee [6] can be used. First, narrative engagement combines emotional intensity with the coherence of the story being told. Sensory-motor engagement is related to the physical presence of the virtual environment and its realism. Finally, motor engagement is achieved through the simplicity and reliability of the provided device. In this study, a smartphone-based player was designed to create an AAR experience which takes into account the listener position as well as head rotations. In addition, spatial audio technologies was used to create accurate virtual acoustics. The present work aims to evaluate the different characteristics of this AAR experience in order to determine how much each one is responsible for the sense of immersion.

2 Methods

2.1 Writing a Geolocated Soundwalk

The soundwalk was created based on excavation data [3] and repeated interviews with the excavation director, Françoise Dumasy. The 25-minute long audio production related the inauguration of the theater in the mid-2nd century AD. A special attention was paid to making the dialogues more engaging, developing the characters and introducing comedic situations to spark the visitor's narrative—and even emotional—interest. About ten zones have been defined within the theater, each representing a part of the story. During the tour, audio files were triggered depending on the listener's

2026. Manuscript submitted to ACM

arrival at GPS locations and the previously visited zones, thanks to an existing web editor developed by Ars Nomadis [9][1] (Figure 1).

2.2 Creating a Virtual Acoustic Environment

The virtual sound environment was created using a third-order ambisonic workflow. Ambisonics is a standard framework for sound field reproduction and is commonly used in extended reality (XR) for its capacity to seamlessly apply rotations to a sound scene [11]. The ancient acoustics at various points in the theater were simulated using a digital model drawn using Blender[2] (see Figure 2). Twenty-four Spatial Room Impulse Responses (SRIR) were calculated using geometrical acoustic algorithms [5] included in the Pyroomacoustics Python package [7, 8]. Each SRIR described the acoustic path between a sound source and the listener, and enabled to create a plausible auralization of the sound space [1]. In addition, some sound sources were meant to be spatialized in the existing environment. To this end, acoustic measurements were made *in situ* using the SpcMic ambisonic microphone[3] to provide SRIR of the existing site.



Fig. 1. Description of geolocated zones in Ars Nomadis Editor.

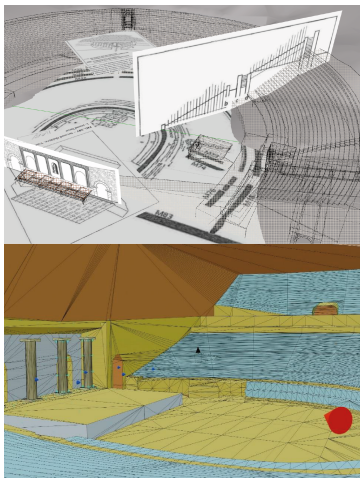


Fig. 2. 3D rendering of the theater based on excavation data (top) and an acoustic simulation of the model (bottom).



Fig. 3. A visitor carrying the technical setup of the soundwalk.

2.3 Designing a Soundwalk Player

Each visitor was equipped with a smartphone and headphones. Smartphones have the advantage of incorporating most of the features needed to build a geolocated audio player: a battery, Global Navigation Satellite System (GNSS), a headphone output, and a USB port. In addition, an Inertial Measurement Unit (IMU) was placed on top on the headphones for head tracking. This sensor allowed to take into account head rotations to accurately spatialize virtual sound sources around the listener (Figure 3). A mobile application was developed by Noise Makers based on the Ars Nomadis tool (2.1) using Unity[4]. To the best of the authors' knowledge, no existing application was able to perform both geolocation and head tracking at the time of development. A single button on the user interface allowed the visitor to start the tour. During the tour, the visitor was guided solely by sound.

[1]https://editor.arsnomadis.eu
[2]https://blender.org
[3]https://spcmic.com/
[4]https://unity.com

3 Results

A questionnaire was given to the 343 participants who experienced the soundwalk. Fifteen closed-ended questions were asked using a 7-point Likert scale. Participants were asked to rate their experience about perceptual sensations on a scale from "0—Disagree" to 6—"Strongly Agree". Results showed that 93% of them would describe this experience as "immersive" and would like to repeat it in a different location. This analysis can be further explored in light of Lee's three criteria [6] described above. To assess the weight of each factor in the sense of immersion, a multiple linear regression (Table 1) was calculated between the statement Y *"I would describe this experience as 'immersive'"* and these three independent variables:

- *I was captivated by the narration of the audio walk* (PC_1, narrative engagement);
- *I was able to forget about the technical setup* (PC_2, motor engagement) ;
- *The virtual sounds seemed to come from the external environment* (PC_3, sensorimotor engagement).

	β	t	p	Standard deviation
	5.034	84.32	$< .001$	0.059
PC_1	0.689	12.30	$< .001$	0.056
PC_2	0.344	6.136	$< .001$	0.056
PC_3	0.231	4.116	$< .001$	0.056

Table 1. Multiple linear regression model associated with the dependent statement Y: *"I would describe this experience as 'immersive'"*. The R^2-coefficient associated is 0.5. The model therefore explains 50% of the variance in the independent variable Y.

This highlights a highly significant relationship between each component and the qualification of an "immersive experience" ($t_i > 2\ \forall i$ and $p_i < 0.001\ \forall i$). Engagement with the narrative is the factor that most influences the sense of immersion ($\beta_{PC1} = 0.689$). Forgetting about the technical setup affects this sensation to half the extent ($\beta_{PC2} = 0.344$). However, it is conceivable that this component would carry more weight with a more discreet device. Furthermore, geolocation approximations hindered the experience for approximately 5% of visitors. Although their impact was not measured, they are not negligible in terms of visitors' sense of letting go—and thus of immersion.

We finally found the parameter for the externalization of sound sources, which may indicate the presence of virtual sources in the real environment ($\beta_{PC3} = 0.231$). A Pearson correlation also reveals a strong relationship between the visitor immersion, the naturalness of the perceived sound sources ($r = 0.54$, $p < 0.001$), and the auralization of the site in ancient times ($r = 0.55$, $p < 0.001$)[5].

4 Conclusion

This project incorporates head tracking and acoustic simulation solutions into an archaeological site tour with a smartphone-based device. Predefined criteria were applied to create this AAR experience and prioritize the parameters contributing to immersion in a virtual audio environment. The accuracy of the narrative appears to be the key factor, followed by the ability to forget the technical setup and the plausibility of virtual sound sources. The interest shown by local spectators in this event is encouraging the deployment of such tools at other venues. The main areas for future work are simplifying the technical setup and improving geolocation technology. Moving away from the smartphone in favor of an embedded device is one of the avenues currently being explored by the authors.

[5]Results obtained from the questions *"The positioning of the sound sources seemed natural to me"* and *"I was able to clearly visualize the site in ancient times"*, which were asked of visitors.

Acknowledgments

The authors would like to thank Françoise Dumasy—director of excavations at the Argentomagus archaeological site—as well as Léna Chariéras, Lubin Laniol, Laurent Laubier, Line Ancel, and Francis Dusserre for their contributions to the creation of the soundwalk. This work was conducted as part of a master's thesis in Sound at the École nationale supérieure Louis-Lumière. The complete methodology and more comprehensive results are described in the thesis untitled: *La ruine antique en Réalité Augmentée Audio — Création sonore immersive sur les vestiges du théâtre gallo-romain d'Argentomagus.*

References

[1] Fabian Brinkmann, Lukas Aspöck, David Ackermann, Steffen Lepa, Michael Vorländer, and Stefan Weinzierl. 2019. A round robin on room acoustical simulation and auralization. *The Journal of the Acoustical Society of America* 145 (2019). doi:10.1121/1.5096178

[2] Suzie Cloves. 2025. Evaluating Geolocated Sound to Interpret and Research Historic Environments: Workflow for a Sonic XR as Outdoor Place History. *Journal on Computing and Cultural Heritage* 18, 2 (2025). doi:10.1145/3731248

[3] Françoise Dumasy. 2000. *Le théâtre d'Argentomagus (Saint-Marcel, Indre).* Éditions de la Maison des sciences de l'homme, Paris.

[4] Matias Harju. 2025. *Audio Augmented Reality: concepts, technologies and narratives.* Routledge, Taylor & Francis Group, London and New York.

[5] Heinrich Kuttruff. 2006. *Room acoustics* (4th ed.). Taylor & Francis, London and New York.

[6] Hyunkook Lee. 2025. A unified conceptual model of immersive experience in extended reality. *Computers in Human Behavior Reports* 18 (2025). doi:10.1016/j.chbr.2025.100663

[7] François Salmon, Charles Verron, Cédric Camier, and Marina Malgrange. 2024. Adaptation and evaluation of Pyroomacoustics for auralization purposes. *INTER-NOISE and NOISE-CON Congress and Conference Proceedings* 270, 6 (2024), 5906–5917. doi:10.3397/IN_2024_3660

[8] R Scheibler, E Bezzam, and I Dokmanic. [n. d.]. Pyroomacoustics: A Python package for audio room simulations and array processing algorithms. arXiv 2018. *arXiv preprint arXiv:1710.04196* ([n. d.]).

[9] Denez Thomas, Antoine Beaufort, and Charles Verron. 2025. Designing a Geolocated Soundwalk Player Using DIY Electronics: An Underestimated Approach to Site-specific Sound Art and Research. In *Proceedings of the 17th International Symposium on Computer Music Multidisciplinary Research (CMMR 2025).* London, 859–870. doi:10.5281/zenodo.17502449

[10] Michael Vorländer. 2008. *Auralization: Fundamentals of Acoustics, Modelling, Simulation, Algorithms and Acoustic Virtual Reality* (1st ed.). Springer, Berlin and Heidelberg.

[11] Franz Zotter and Matthias Frank. 2019. *Ambisonics: A Practical 3D Audio Theory for Recording, Studio Production, Sound Reinforcement, and Virtual Reality.* Springer International Publishing, Cham. doi:10.1007/978-3-030-17207-7